

Worksheet 0a R – creating data frames and reading data from files

Age	Race	Height	Income	IsMale	Politics
22	1	Tall	0.39	TRUE	moderate
33	3	Short	0.34	TRUE	liberal
52	1	Moderate	0.51	FALSE	moderate
46	6	Tall	0.63	TRUE	conservative

Modified from source: <https://jamesmccaffrey.wordpress.com/2020/04/21/example-of-calculating-the-gower-distance/>

(a) For each variable above, identify if it is a categorical variable or a quantitative one. If it is a categorical variable, further classify the variable as nominal (or binary) or ordinal.

(b) Use R to create each variable, making sure the type matches your answer in (a). Then create a data frame in R to collectively house these variables as a data set.

- (c) Type the data into a text file. Use `read.table()` to read the contents of the text file. Is the output of `read.table()` a data frame or some other data structure? Do the variables match the type in part (a)? If not, how do you convert them?
- (d) Type the data into an Excel file. Use `read_excel()` in package “readxl” to read the contents of the Excel file. Is the output a data frame or some other data structure? If it is not a data frame, how do you convert it into a data frame? Do the variables match the type in part (a)?
- (e) Create a CSV file using the data. Use `read.csv()` to read the contents of the CSV file. Is the output a data frame or some other data structure? Do the variables match the type in part (a)?